Career Launcher MBA

QA Modern Mathematics-03

Probability

Question 1

If 10 cars are parked in a row, then what is the probability that there are exactly 5 cars between 2 particular cars?

(1) $\frac{8 \times 5!}{10!}$

(2) $\frac{4 \times 5! \times 3! \times 2!}{10!}$

(3) $\frac{8 \times 8!}{10!}$

(4) $\frac{16 \times 8!}{10!}$

Question 2

Three shots are fired at a target. The odds in favour of the first hitting the target is 2 : 1, that of the second hitting the target is 3 : 1 and the odds against the third hitting the target is 2 : 1. What is the probability of at least 2 shots hitting the target?

(1) 1/3

(2) 13/36

(3) 19/36

(4) 23/36

Question 3

If you draw 4 playing cards at random, one by one without replacement from a well-shuffled pack of 52 cards, what is the probability that they will be of different suits as well as different denominations ?

(1) $\frac{4}{{}^{13}C_1}$

(2) $\frac{4}{{}^{52}C_4}$

(3) $\frac{4}{{}^{52}C_2}$

(4) $\frac{52 \times 36 \times 22 \times 10}{52 \times 51 \times 50 \times 49}$

Question 4

In a throw of 13 unbiased dice, the most probable sum(s) of the numbers that show on the upper surfaces of dice is

(1) 45

(2) 46

(3) 39

(4) Either (1) or (2)

Question 5

If you pick up 2 numbers out of first 29 multiples of 11, then what is the probability that the product of these 2 numbers is 6534?

(1) $\frac{8}{{}^{29}C_2}$

(2) $\frac{4}{{}^{29}C_2}$

(3) $\frac{3}{{}^{29}C_2}$

(4) $\frac{6}{{}^{29}C_2}$

Question 6

On any day, Raman is twice as likely to go to gymnasium, as he is likely to go for volleyball practice. Given that Raman necessarily goes for exactly one of the two activities each day, what is the probability that he goes to the gymnasium on 5 days out of 7 consecutive days of a specified week?

(1) $\frac{32}{2187}$

(2) $\frac{28}{729}$

(3) $\frac{4}{2187}$

(4) $\frac{224}{729}$

Question 7

Three integers are randomly chosen without replacement from the first 20 positive integers. Find the probability that these three integers form an arithmetic progression.

(1) $\frac{5}{38}$

(2) $\frac{5}{57}$

(3) $\frac{3}{38}$

(4) $\frac{1}{19}$

Question 8

Three vertices of a regular hexagon are chosen at random. The probability that the triangle with these vertices is not equilateral, equals

(1) 1/10

(2) 9/10

(3) 1/2

(4) 3/4

Question 9

A and B toss a fair coin 50 times each simultaneously. What is the probability that both of them will not get tails at the same toss?

(1) $\left(\frac{1}{4}\right)^{50}$

(2) $\left(\frac{2}{3}\right)^{50}$

(3) $\left(\frac{1}{2}\right)^{50}$

(4) $\left(\frac{3}{4}\right)^{50}$

Question 10

P and Q are playing a game in which they toss a biased coin alternately. The probability that the coin shows a head is 0.2. The game continues until the coin shows head and the person on whose turn a head shows wins the game. Find the probability that P wins the game. (Assume that Q tosses the coin first)

(1) 1/3

(2) 1/2

(3) 2/3

(4) 4/9

Question 11

John plans to order an anniversary gift for his friend from an online retailer. However, the anniversary coincides with the festival season during which there is a huge demand for buying online goods and hence deliveries are often delayed. He estimates that the probability of receiving the gift, in time, from the retailers P, Q, R and S are 0.7, 0.8, 0.9 and 0.5 respectively. He orders from all four retailers simultaneously. What would be the probability that his friend would receive the gift in time?

(1) 0.004

(2) 0.006

(3) 0.216

(4) 0.997

Question 12

Rahul has a total of 18 red and black pens in two boxes (each box has pens of both colors). A pen is randomly drawn from the first box followed by another randomly drawn from the second box, the probability of both being red is 5/16. What is the probability of both pens being black?

(1) 1/16

(2) 2/16

(3) 3/16

(4) 4/16

Question 13

The scheduling officer for a local police department is trying to schedule additional patrol units in each of two neighbourhoods – southern and northern. She knows that on any given day, the probabilities of major crimes and minor crimes being committed in the northern neighbourhood were 0.418 and 0.612, respectively, and that the corresponding probabilities in the southern neighbourhood were 0.512 and 0.520. Assuming that all crime occur independent of each other and likewise that crime in the two neighbourhoods are independent of each other, what is the probability that no crime of either type is committed in either neighbourhood on any given day?

(1) 0.069

(2) 0.225

(3) 0.052

(4) 0.775

Question 14

There are four machines in a factory. At exactly 8 pm, when the mechanic is about to leave the factory, he is informed that two of the four machines are not working properly. The mechanic is in a hurry, and decides that he will identify the two faulty machines before going home, and repair them next morning. It takes him twenty minutes to walk to the bus stop. The last bus leaves at 8 : 32 pm. If it takes six minutes to identify whether a machine is defective or not, and if he decides to check the machines at random, what is the probability that the mechanic will be able to catch the last bus?

(1) 0

(2) 1/6

(3) 1/4

(4) 1/3

Question 15

The supervisor of a packaging unit of a milk plant is being pressurised to finish the job closer to the distribution time, thus giving the production staff more leeway to cater to last minute demand. He has the option of running the unit at normal speed or at 110% of normal - “fast speed”. He produces 60% of packets at the fast speed and the rest at the normal speed. A packet produced at the fast speed is twice as likely to be damaged as compared to a packet produced at the normal speed. If the probability of randomly selected packet from the unit being damaged is 0.096, what is the probability that the packet will not be damaged at normal speed?

(1) 0.81

(2) 0.94

(3) 0.75

(4) 0.60

Thanks!